AI and ML for Architecture (M25RD204)

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Course Contents & Schedule

1. Python Programming + Relevant Libraries : 1.7.2026
2. Introduction to AI, Data Science and ML – Data Collection and Preprocessing : 2.7.2026
3. Supervised Learning Algorithms – Linear Regression for Space Planning : 2.7.2026
4. Unsupervised Learning – K - Means for Site Analysis : 2.7.2026
5. Neural Networks, Deep Learning : 6.7.2026
6. Computer Vision – CNNS, Image Classification, Stable Diffusion, Mid Journey : 6.7.2026
7. Generative Design with GANs : 7.7.2026
8. Generative AI – LLMs, VLMs, Multimodal AI Models : 7.7.2026
9. Generative AI Tools for Architecture : 7.7.2026

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10. Natural Language Processing for Design Briefs – Text Analysis and Brief Processing : 8.7.2026
11. Reinforcement Learning for Space Utilization : 8.7.2026
12. Optimization Algorithms – Genetic Algorithm for Layout Planning : 9.7.2026
13. Building Performance Prediction – Energy Consumption ML Models : 13.7.2026
14. Smart Building Systems – IoT Integration with ML Planning : 13.7.2026
15. Advanced Applications - Custom AI Tool Development : 14.7.2026
16. AI Ethics in Architecture – Bias Detection and Mitigation : 14.7.2026
17. Industry Case Studies – Portfolio Compilation: 14.7.2026
18. Project Presentations – 15.7.2026

Artificial Intelligence

- **Artificial Intelligence (AI)** is a field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence. These tasks include reasoning, learning, problem-solving, understanding language, and perceiving environments.
- AI is an umbrella term that covers two distinct methodologies:
  - Deterministic AI (Non-ML)
  - Probabilistic AI (Machine Learning)

Deterministic Methods for building AI

1. Rule-Based & Expert Systems
2. Symbolic AI (Good Old Fashioned AI)
3. Search & Planning Algorithms
4. Fuzzy Logic
5. Probabilistic Programming

Machine Learning

- **Machine Learning (ML)** is a subset of artificial intelligence (AI) that allows computers to learn and make decisions without being explicitly programmed.

- Instead of a human writing rigid rules, ML algorithms analyze vast datasets, identify hidden patterns, and build statistical models to solve complex problems on their own.

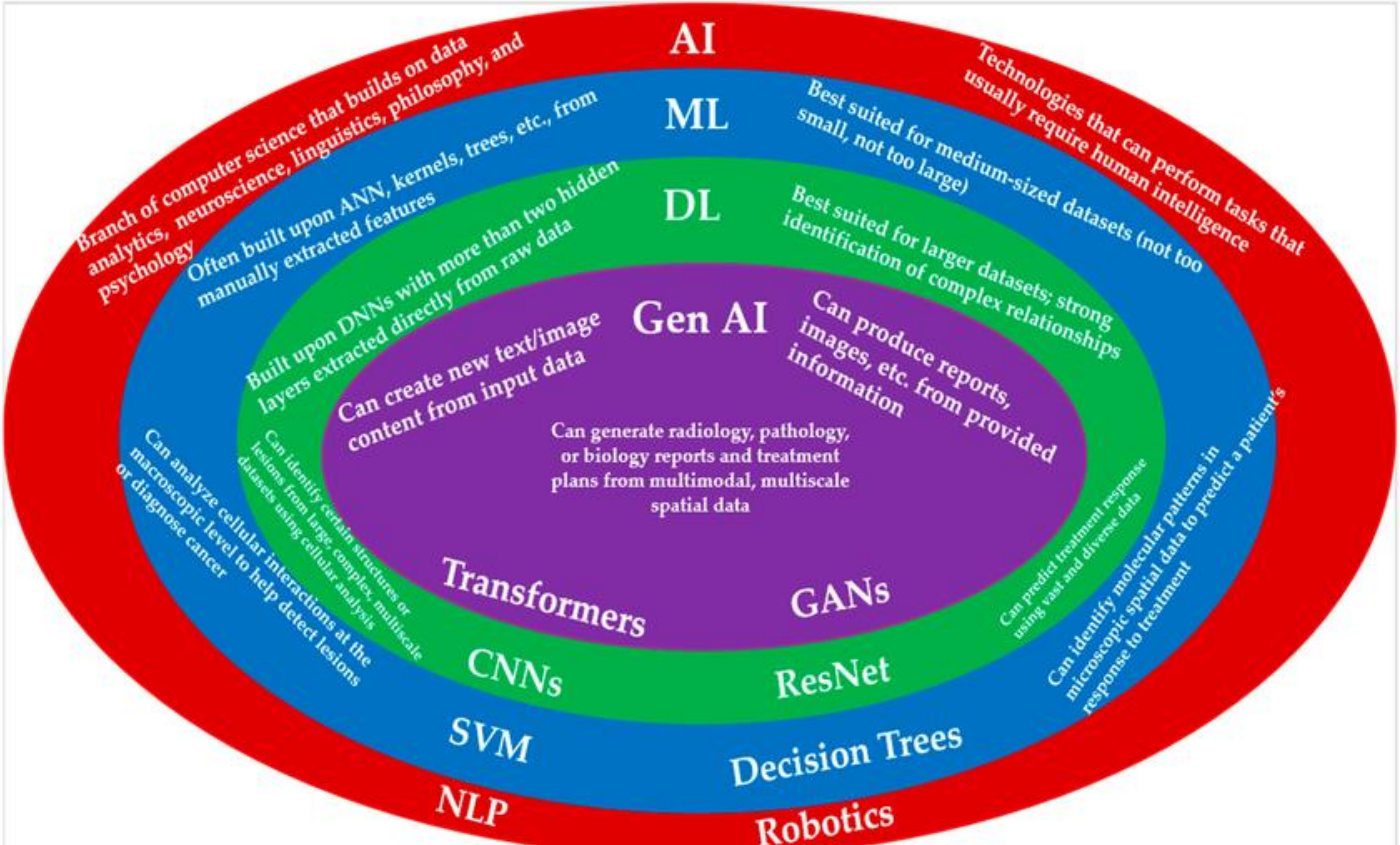
How Machine Learning Works

The standard ML process follows four major phases:

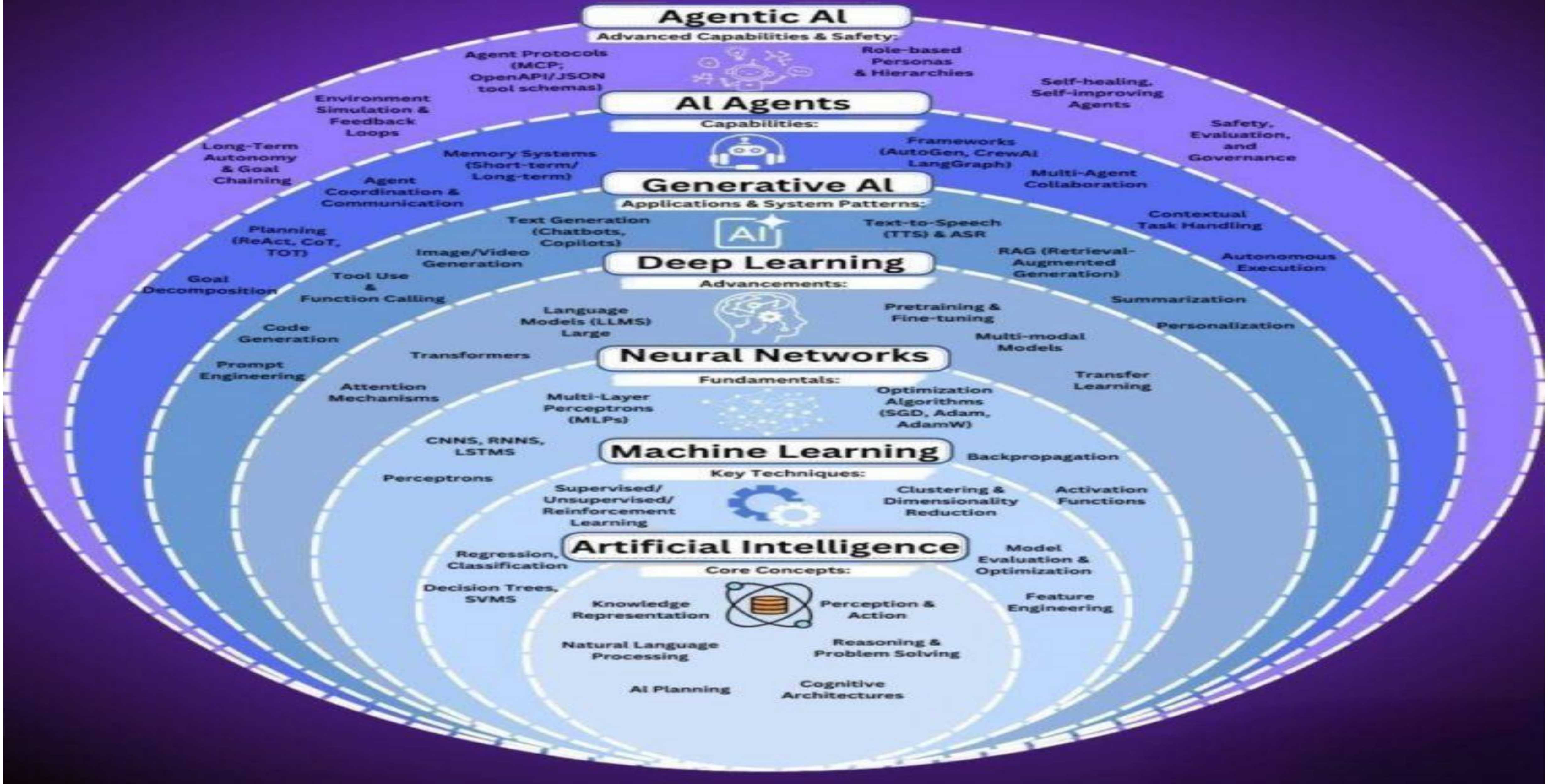
- **Data Collection:** Gathering and cleaning large amounts of relevant information.
- **Model Training:** Feeding data into an algorithm to find patterns and mathematical relationships.
- **Testing:** Evaluating the model using new data to check its prediction accuracy.
- **Inference:** Deploying the finalized model to make predictions on real-world inputs.

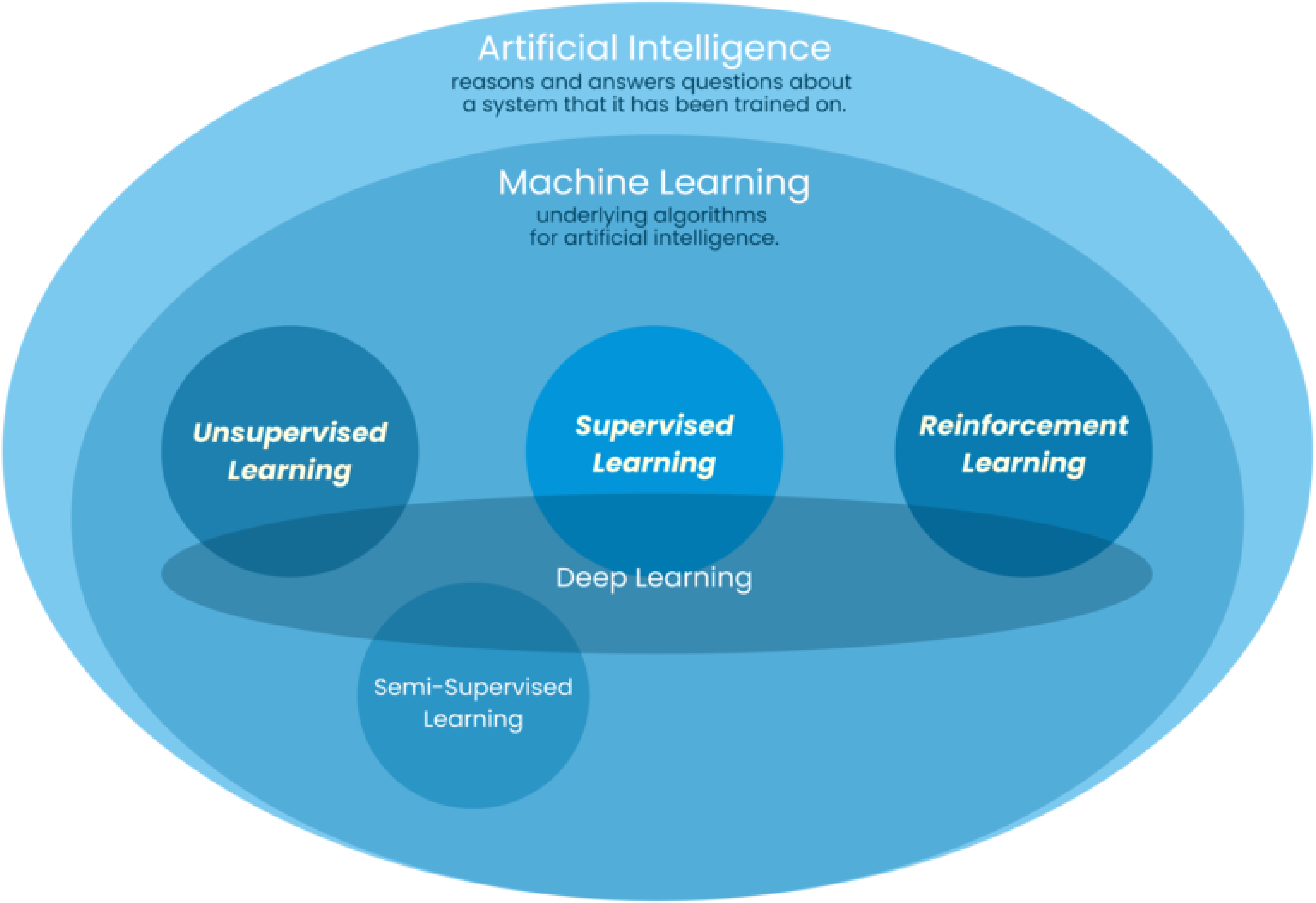
The 4 Main Types of Machine Learning

- **Supervised Learning:** The algorithm trains on labelled data (input data paired with the correct output). For example, teaching a system to identify spam by showing it millions of emails marked "spam" or "not spam”
- **Unsupervised Learning:** The algorithm analyses unlabelled data to find hidden structures or groupings on its own. For example, clustering e-commerce customers into different segments based on purchasing habits.
- **Semi-Supervised Learning:** The system trains on a small amount of labelled data combined with a massive amount of unlabelled data. This balances high accuracy with lower data - labelling costs.
- **Reinforcement Learning:** The algorithm learns through trial and error, receiving virtual rewards for correct actions and penalties for mistakes. This method is heavily used to train robotics and self-driving cars.


AI
Branch of computer science that builds on data analytics, neuroscience, linguistics, philosophy, and psychology
Technologies that can perform tasks that usually require human intelligence
ML
Often built upon ANN, kernels, trees, etc., from manually extracted features
Best suited for medium-sized datasets (not too small, not too large)
DL
Built upon DNNs with more than two hidden layers extracted directly from raw data
Best suited for larger datasets; strong identification of complex relationships
Gen AI
Can create new text/image content from input data
Can produce reports, images, etc. from provided information
Can generate radiology, pathology, or biology reports and treatment plans from multimodal, multiscale spatial data
Can analyze cellular interactions at the macroscopic level to help detect lesions or diagnose cancer
Can identify certain structures or lesions from large, complex, multiscale datasets using cellular analysis
Can predict treatment response using vast and diverse data
Can identify molecular patterns in microscopic spatial data to predict a patient's response to treatment
Transformers
GANs
CNNs
ResNet
SVM
Decision Trees
NLP
Robotics

The Agentic AI Universe



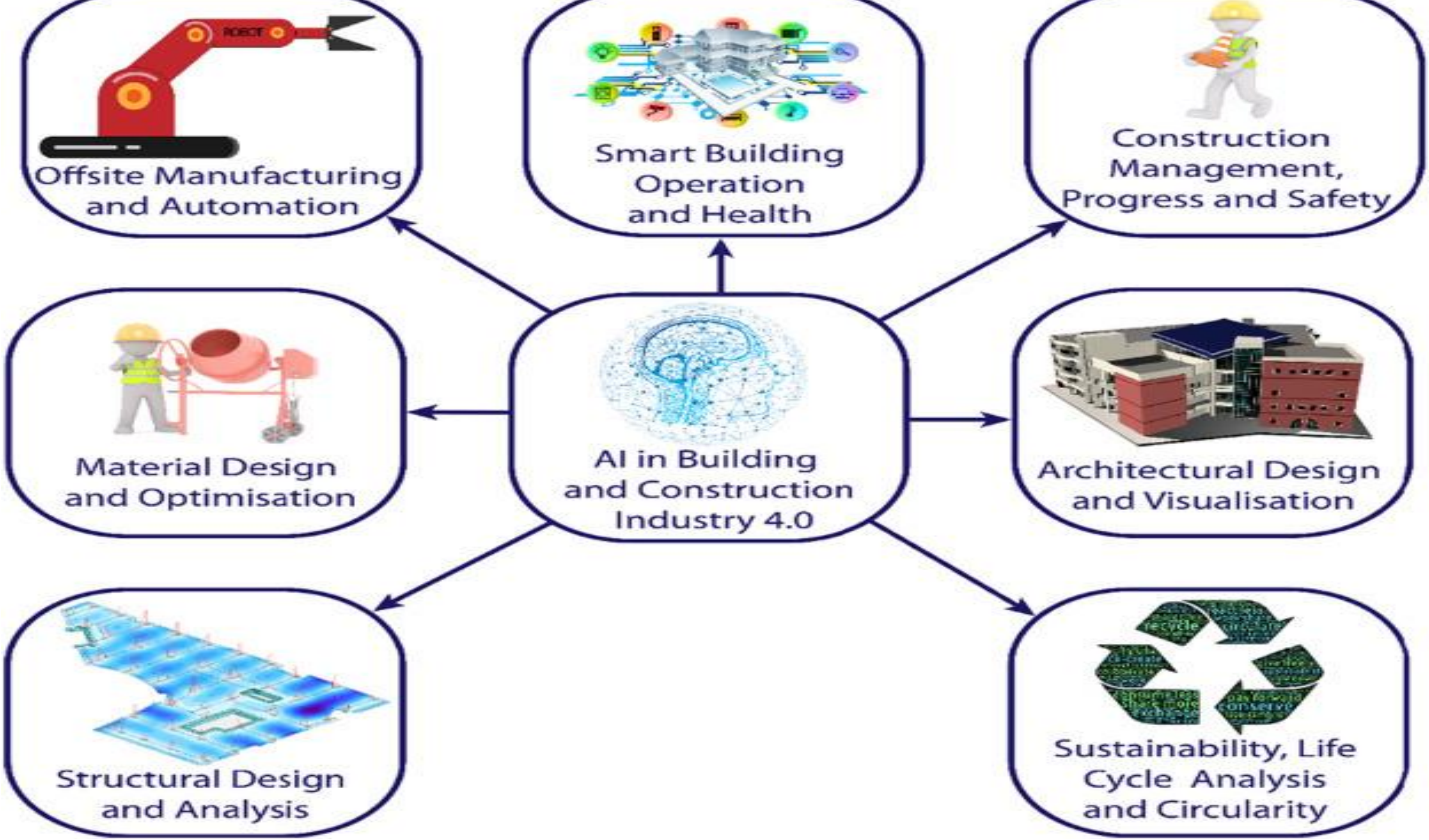

Artificial Intelligence
reasons and answers questions about a system that it has been trained on.
Machine Learning
underlying algorithms for artificial intelligence.
Unsupervised Learning
Supervised Learning
Reinforcement Learning
Deep Learning
Semi-Supervised Learning

Capabilities of AI

Modern AI systems generally excel at four core capabilities: [1]

- **Perception:** Translating real-world inputs (audio, video, text) into digital data.
- **Reasoning:** Using logic or statistical probability to solve complex problems.
- **Learning:** Improving performance over time based on new data or feedback.
- **Action:** Making decisions, generating content, or triggering physical movements


AI-driven optimization
ML-Tools
ML-predictive maintenance
AI
AI-driven optimization
Predictive maintenance
BIM
Facility management


Offsite Manufacturing and Automation
Smart Building Operation and Health
Construction Management, Progress and Safety
Material Design and Optimisation
AI in Building and Construction Industry 4.0
Architectural Design and Visualisation
Structural Design and Analysis
Sustainability, Life Cycle Analysis and Circularity

The Two Categories of AI

- **Narrow AI (Weak AI):** Systems designed to handle one specific task extremely well. Examples include search engines, voice assistants, and self-driving cars. This is the only form of AI that exists today.

- **General AI (Strong AI / AGI):** A theoretical system with human-like cognitive abilities across all domains. It could understand, learn, and apply knowledge to any problem without specialized training.

Role of Generative AI in Computational Design

Image-to-Image (Img2Img) and Massing Transmutation : You can use Image to Image inputs to take a crude, blocky 3D massing model from Rhino and use AI to generate the architectural detailing, materiality, and lighting over that exact geometry.

- [Crude Rhino 3D Blocks] —-> [AI Processing + Prompt] —-> [Photo-Real Architectural Rendering]
- Automating the tedious task of texture mapping and scene rendering during early client presentations.

Role of Generative AI in Computational Design

Dematerializing the "Technical Wall"

- You don’t need to be an expert in understanding vector math and syntax to see a complex form. Generative AI allows for **semantic design**—using natural language to describe spatial complexity.

- This democratizes the exploration of highly intricate, non-linear geometries (like Baroque detailing or complex organic lattices) for students who haven't yet mastered complex NURBS modelling in Rhino.

Role of Generative AI in Computational Design

Semantic vs. Explicit Geometry: Generative AI generates pixels, not data.

Attribute	**Generative AI (e.g., Midjourney)**	**Computational Design (e.g., Rhino/Grasshopper)**
Output Type	2D Pixel Grid (Images)	3D Mathematical Data (NURBS, Coordinates)
Precision	Probabilistic (Approximated, "Looks right")	Deterministic (Exact, "Is right")
Downstream Use	Client Inspiration, Conceptual Ideation	Fabrication, CNC Milling, Structural Analysis, BIM

Midjourney can dream up a breathtaking, highly complex structural column, but it cannot tell you how thick the steel must be, it cannot export a 3D file to a robotic arm, and it cannot calculate the surface area for cost estimation.

Using both Gen AI and Conventional Design Software in a modern Pipeline

- **Ideation:** Use **Midjourney** to rapidly discover a unique aesthetic language, material palette, or formal silhouette.
- **Translation:** Take that AI-generated inspiration image into **Rhino/Grasshopper**, using Python or visual scripting to decode the underlying geometric logic into actual 3D points and parametric vectors.
- **Execution:** Optimize that 3D model for structural engineering and digital fabrication.

AI is a Tool, Not a Replacement

- Gen AI does not replace architects and designers, it acts as a powerful front-end catalyst, while engines like Rhino 8 remain the mandatory backbone for turning those digital dreams into buildable reality.